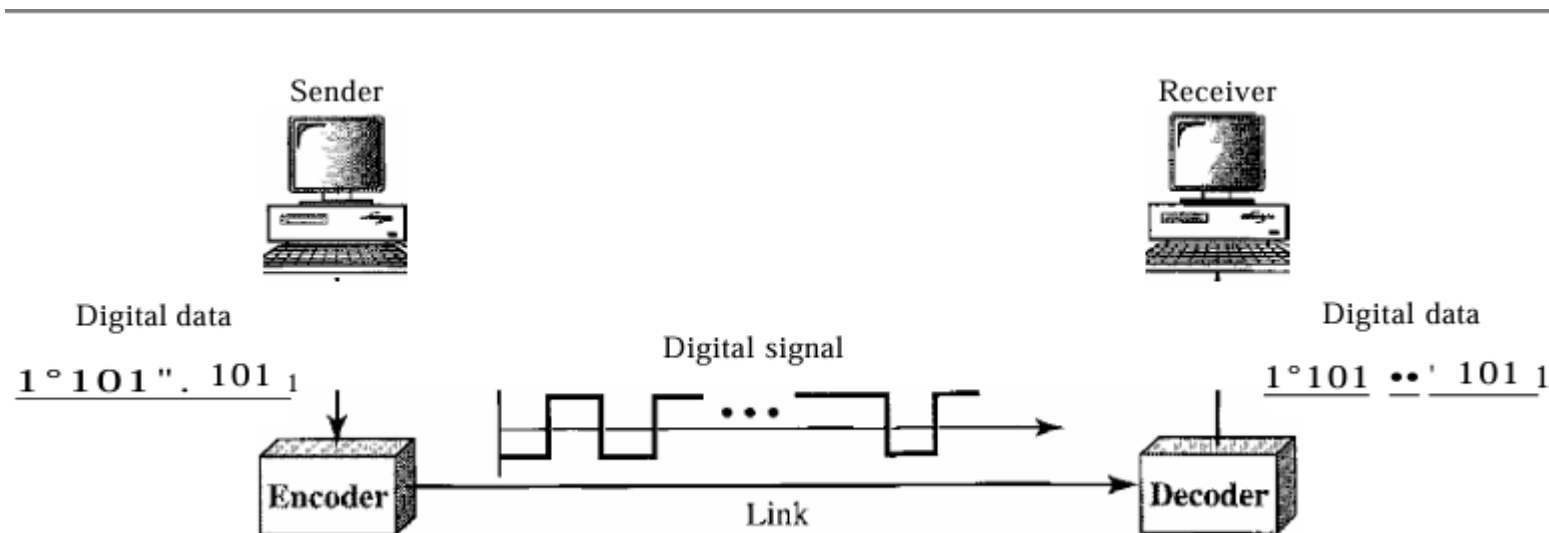


DIGITAL-TO-DIGITAL CONVERSION

- In this section, we see how we can represent digital data by using digital signals.
- The conversion involves three techniques: line coding, block coding, and scrambling.
- **Line coding is always needed;** block coding and scrambling may or may not be. Needed

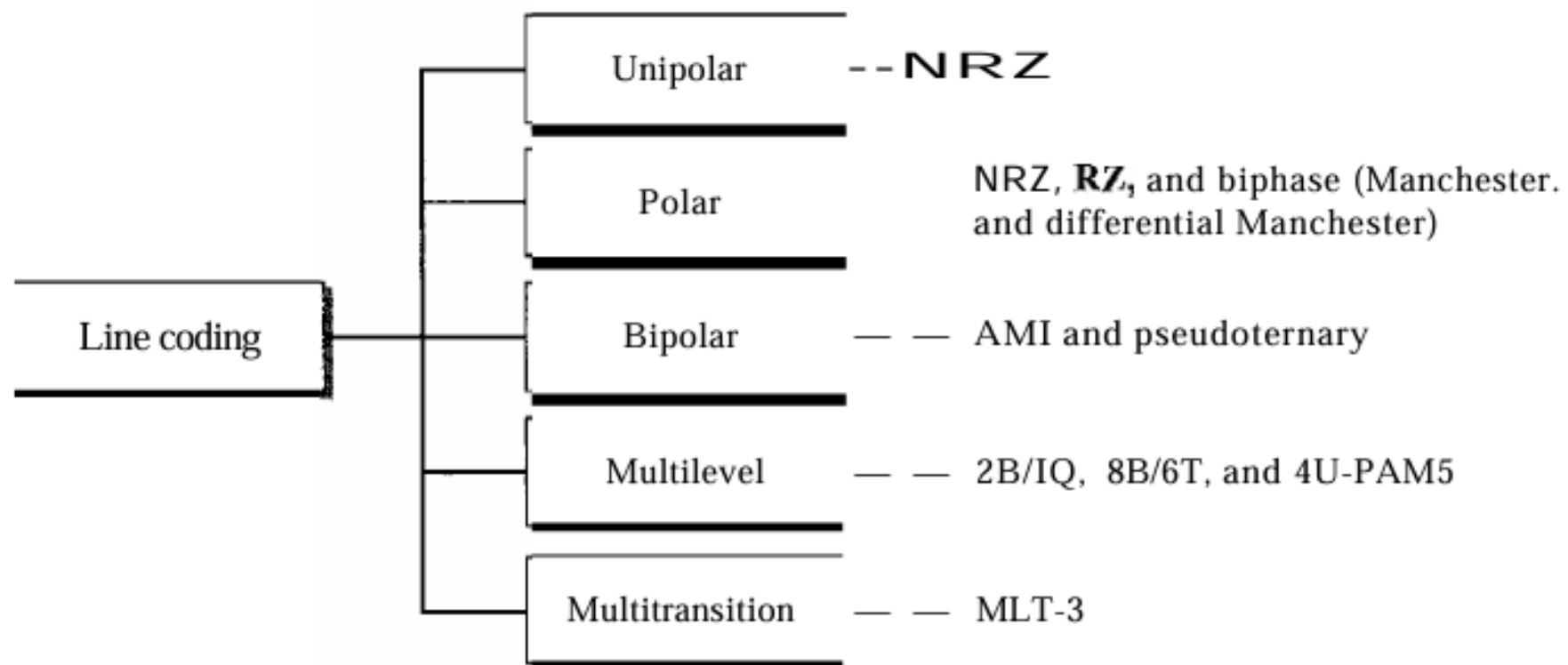
DIGITAL-TO-DIGITAL CONVERSION

- Line coding is the process of converting digital data to digital signals. We assume that data, in the form of text, numbers, graphical images, audio, or video, are stored in computer memory as sequences of bits.



DIGITAL-TO-DIGITAL CONVERSION

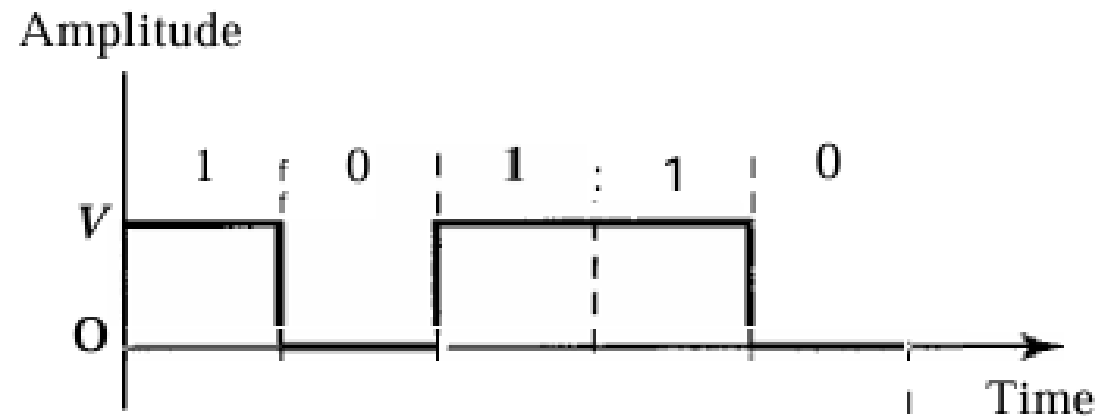
- Line coding scheme



DIGITAL-TO-DIGITAL CONVERSION

- **Unipolar NRZ (Non-Return-to-Zero) scheme**

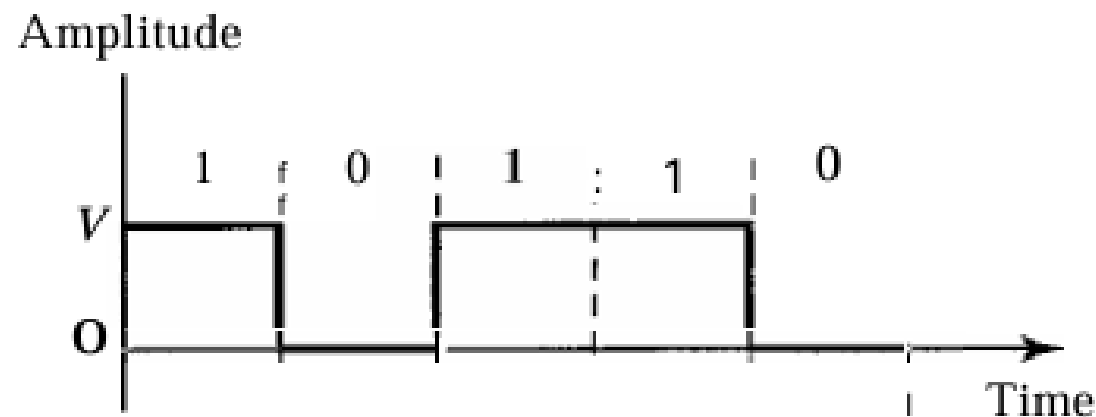
A unipolar scheme was designed as a non-return-to-zero (NRZ) scheme in which the positive voltage defines bit 1 and the zero voltage defines bit 0. It is called NRZ because the signal does not return to zero at the middle of the bit.



DIGITAL-TO-DIGITAL CONVERSION

- **Polar NRZ-L and NRZ-I schemes**

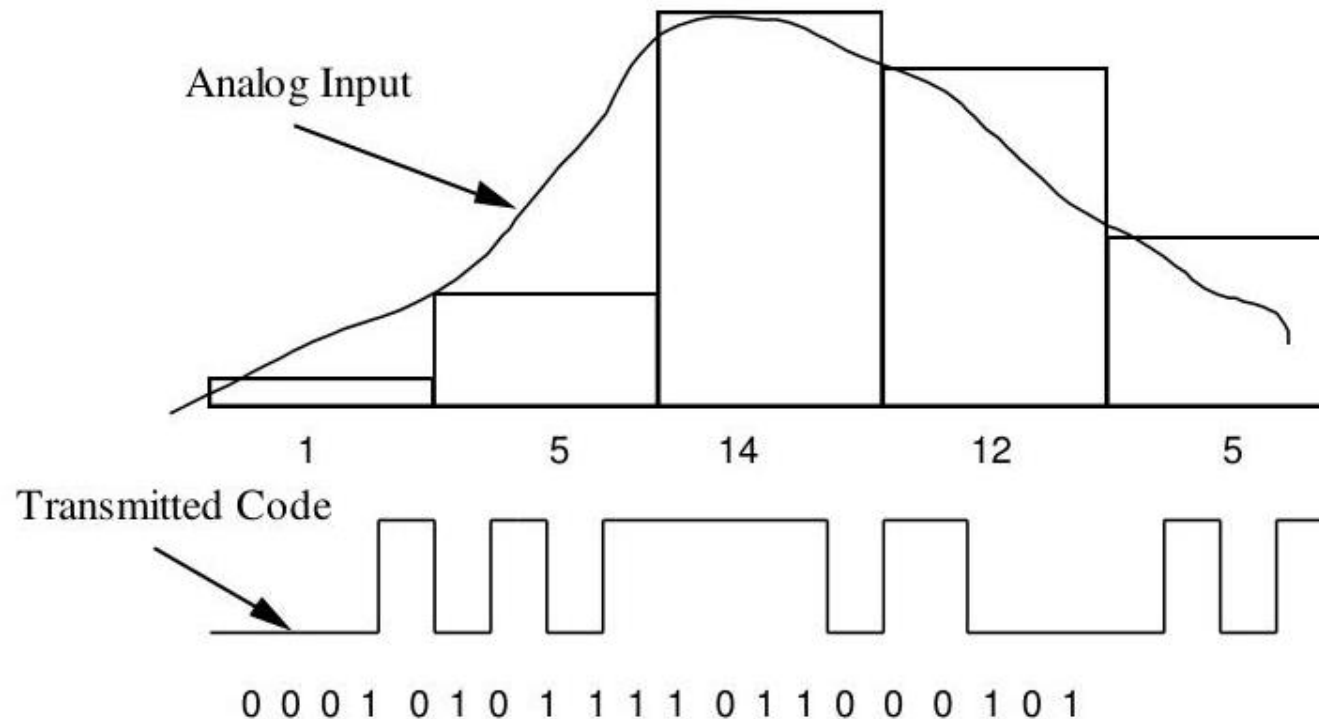
- ❑ In polar schemes, the voltages are on the both sides of the time axis. For example, the voltage level for 0 can be positive and the voltage level for 1 can be negative.
- ❑ In NRZ-L the level of the voltage determines the value of the bit. In NRZ-I the inversion or the lack of inversion determines the value of the bit.



Pulse Code Modulation

Pulse Code Modulation

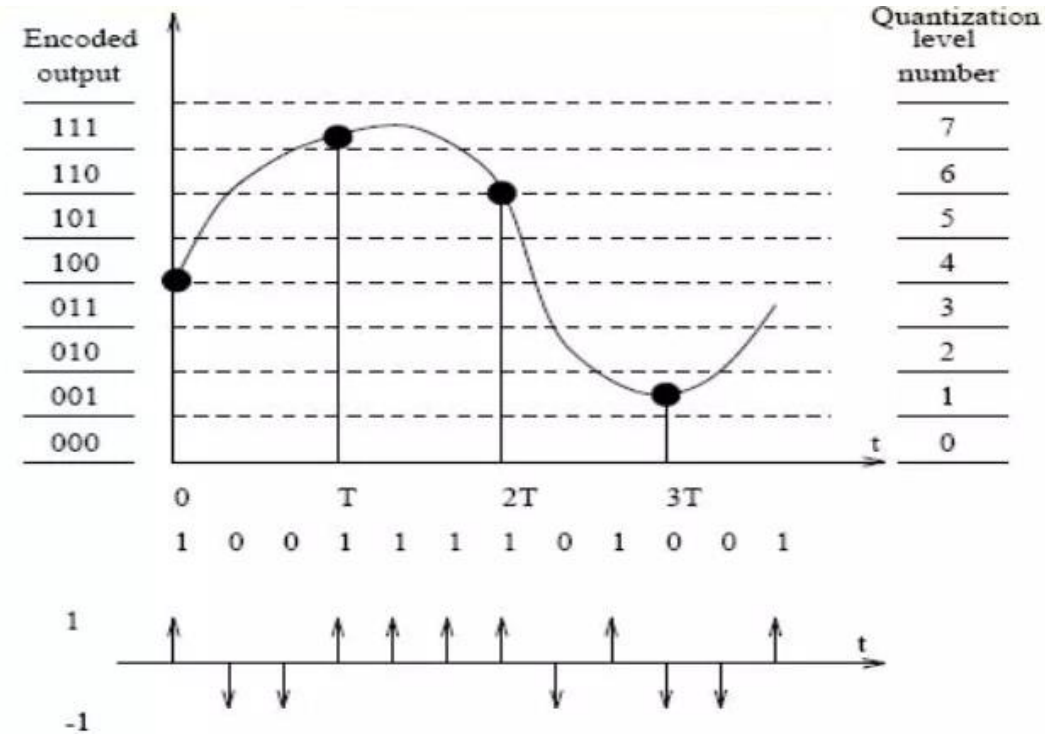
Definition: Pulse code Modulation (PCM) is essentially analog to digital conversion of a special type where the information contained in the instantaneous samples of an analog signal is represented by digital word in a serial bit streams.



Pulse Code Modulation

The PCM signal is generated by carrying out 3 basic operations:

1. Sampling 2. Quantizing 3. Encoding.



□ Each sample value from the analog signal can be any one of an infinite number of levels. So, instead of using exact sample value, the sample is replaced by the closest allowed value.

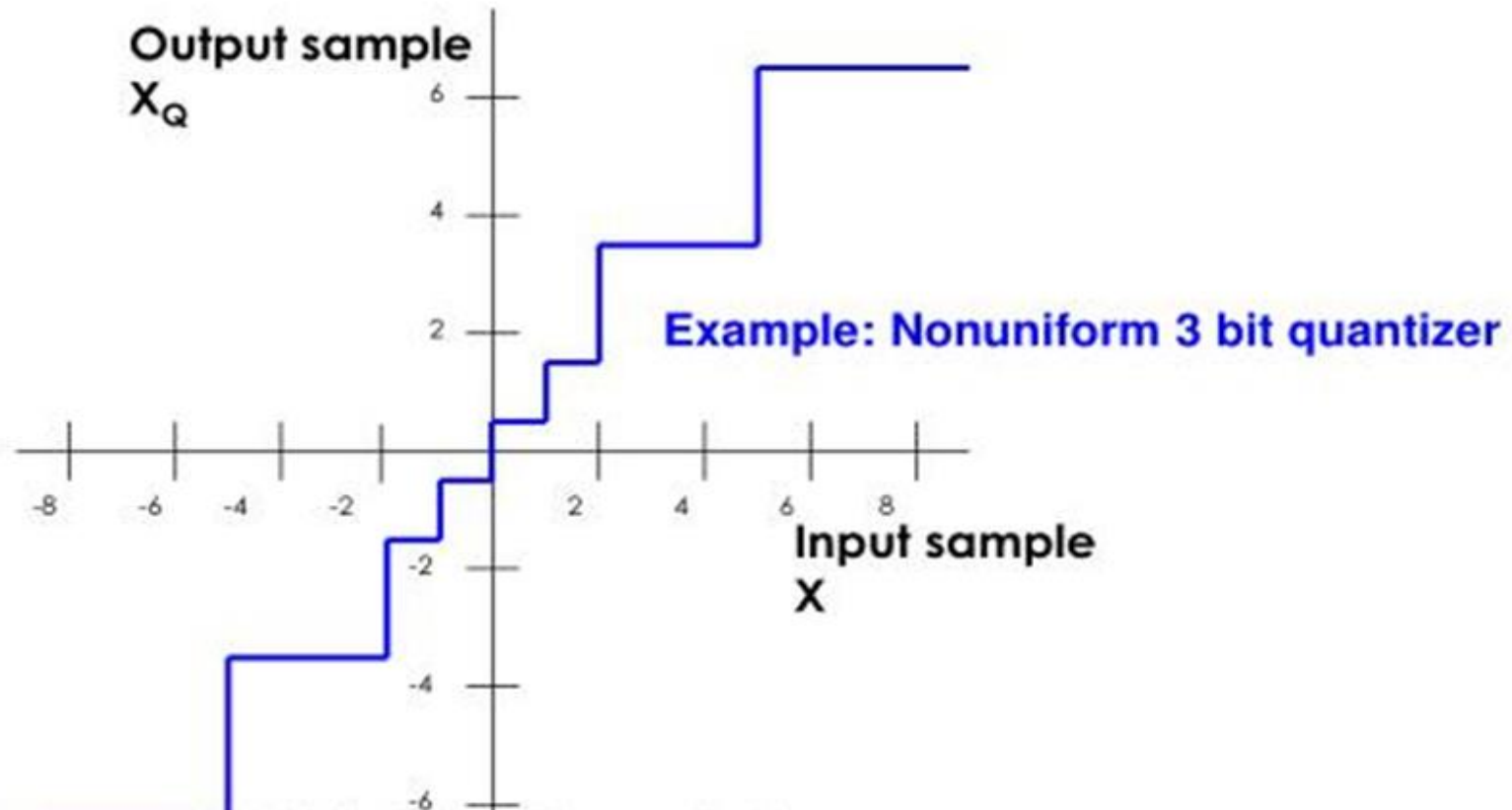
□ Then for encoding binary code, or any other code may be used.

- SNR is an indication of the quality of the received signal
- Ideally we would like to have constant SNR
- Unfortunately, the SNR is directly proportional to the signal power, which varies from talker to talker
- The signal power can also vary because of the connecting circuits
- SNR vary even for the same talker, when the person speaks softly
- Smaller amplitudes pre-dominate in speech and larger amplitude much less frequent.
- This means the SNR will be low most of the time

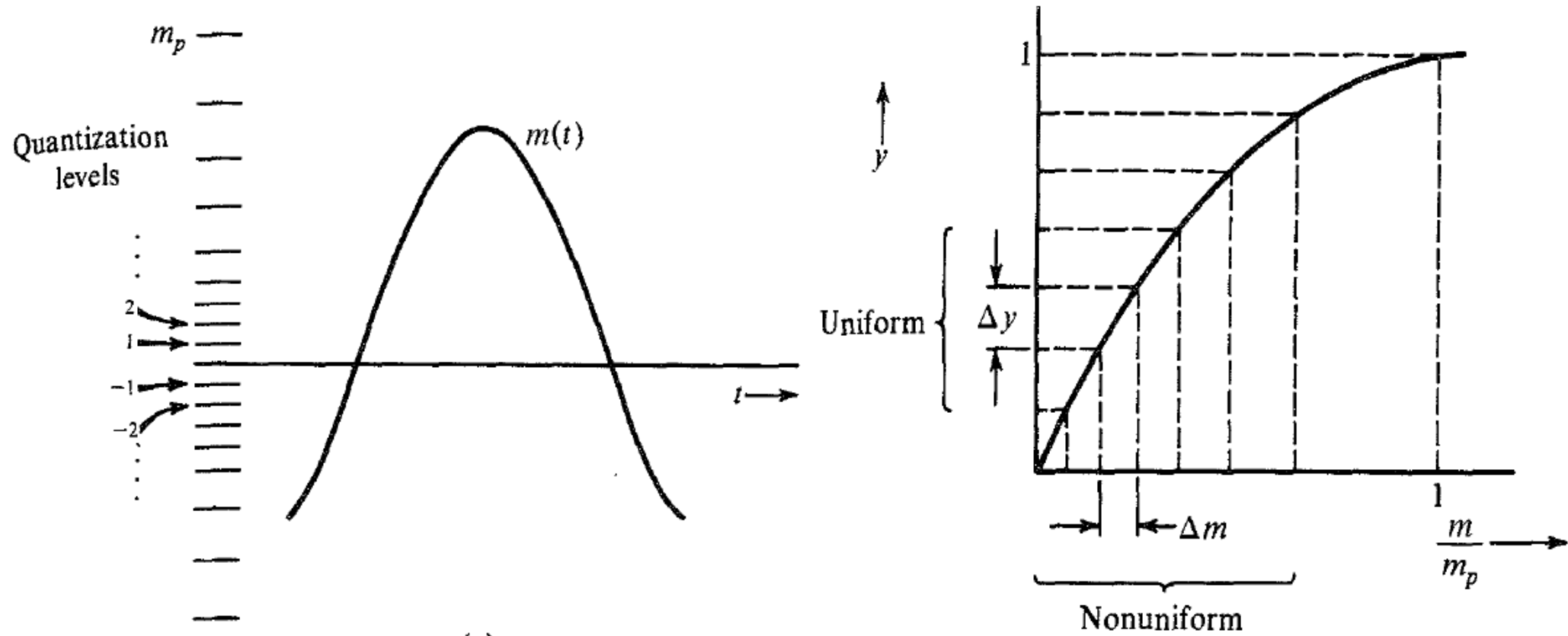
This problem can be solved by non-uniform distribution

Non-Uniform Distribution

- Many signals such as speech have a nonuniform distribution.
 - ▶ The amplitude is more likely to be close to zero than to be at higher levels.
- Nonuniform quantizers have unequally spaced levels
 - ▶ The spacing can be chosen to optimize the SNR for a particular type of signal.



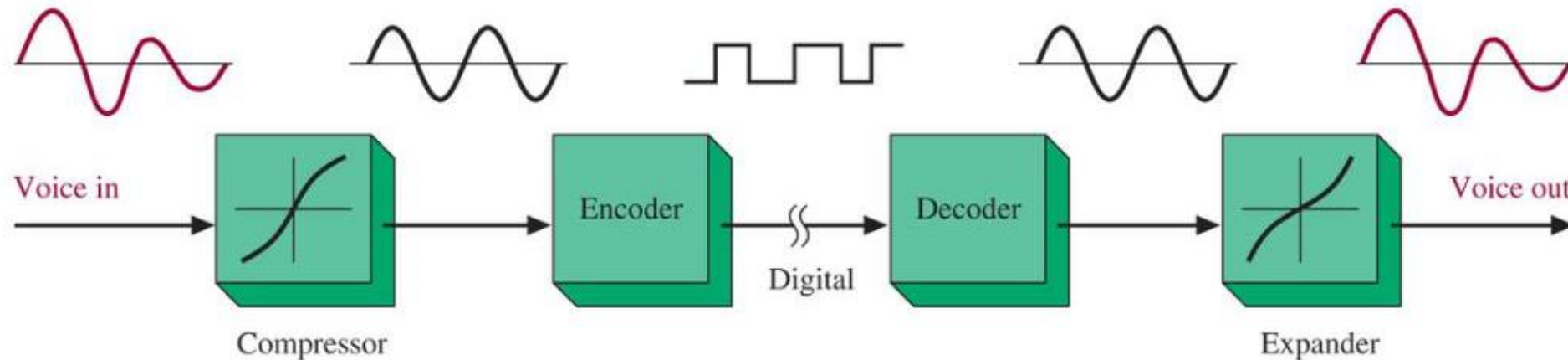
Non-Uniform Distribution



Non uniform distribution

Compressor

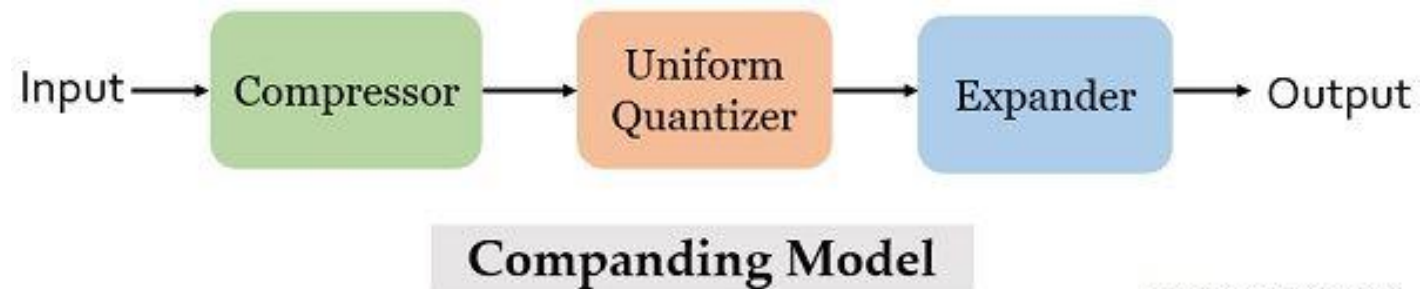
- Nonuniform quantizers are difficult to make and expensive.
- An alternative is to first pass the speech signal through a nonlinearity before quantizing with a uniform quantizer.
- The nonlinearity causes the signal amplitude to be *Compressed*.
 - The input to the quantizer will have a more uniform distribution.
- At the receiver, the signal is *Expanded* by an inverse to the nonlinearity.
- The process of compressing and expanding is called *Comanding*.
- The electronic circuit responsible for comanding is called the *Comandor*.



Compressor

Comanding refers to a technique for compressing and then expanding (or decompressing) an analog or digital signal. It is a combination of the words “compressing” and “expanding.”

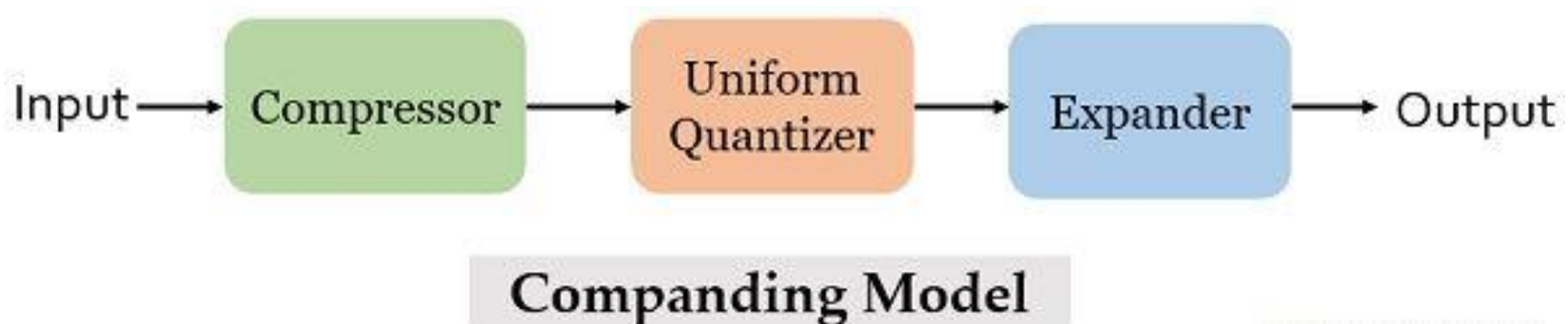
This twin-sequential process is non-linear overall but linear over short periods of time. Data is compressed before being transmitted. Then, it is expanded at the receiving end using the same non-linear scale to restore it to its original form, but with reduced noise and crosstalk levels.



Compendor

Componding refers to a technique for compressing and then expanding (or decompressing) an analog or digital signal. It is a combination of the words “compressing” and “expanding.”

This twin-sequential process is non-linear overall but linear over short periods of time. Data is compressed before being transmitted. Then, it is expanded at the receiving end using the same non-linear scale to restore it to its original form, but with reduced noise and crosstalk levels.



Advantages of PCM:

- *Relatively inexpensive*
- *Easily multiplexed*: PCM waveforms from different sources can be transmitted over a common digital channel (TDM)
- *Easily regenerated*: useful for long-distance communication, e.g. telephone
- Better noise performance than analog system
- Signals may be stored and time-scaled efficiently (e.g., satellite communication)
- Efficient codes are readily available

Disadvantage:

- Requires wider bandwidth than analog signals

For a binary PCM of bandwidth 3 KHz, 8 binary digits are assigned. Sampling frequency $33\frac{1}{3}\%$ over the Nyquist rate. Calculate bandwidth of channel.

Soln: $B = 3\text{ kHz}$

Nyquist, $R_N = 2 \times 3000 = 6000\text{ Hz}$.

$$\text{Actual, } R_A = 6000 \times \left(1\frac{1}{3}\right) = 8000\text{ Hz}$$

$$n = \log_2 256 = 8\text{ bits per sample.}$$

$$\text{total of } C = 8 \times 8000 = 64,000\text{ bit/s.}$$

Because we can transmit up to 2 bit/s per hertz
of bandwidth,

a) Min transmission bandwidth,

$$B_T = C/2 = 32\text{ kHz.}$$